

Adaptive Phosphene Encoding via Learned Stimulus Optimisation for Cortical Visual Prosthetics

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Abstract

Cortical visual prosthetics restore partial sight by electrically stimulating primary visual cortex (V1), producing localised perceptions called phosphenes. Classical encoders map electrode activations directly from image brightness — a hand-crafted heuristic that cannot adapt to the complex, non-linear relationship between stimulation patterns and perceived images. This paper presents a learned adaptive encoder that replaces brightness mapping with a convolutional neural network trained end-to-end through a frozen biologically plausible V1 simulator and a domain-aligned perceptual oracle. Training proceeds in two phases: first, an EfficientNet-B0 oracle is pretrained on phosphene-simulated images to close the domain gap between natural photographs and cortical phosphenes; second, the encoder learns electrode activation patterns by maximising classification accuracy through the frozen simulator and oracle pipeline. On the full CIFAR-10 benchmark the adaptive encoder achieves 61.0% classification accuracy versus 46.4% for brightness mapping — a statistically robust improvement of +14.6 percentage points across all 10 object categories. Ablation experiments confirm that the perceptual loss drives the result, while sparsity and smoothness regularisers reduce mean electrode activation from 47.1% to 28.3%, lowering charge injection independently of classification performance. An electrode budget experiment shows that the adaptive encoder with 50 electrodes (57.2%) outperforms brightness mapping with 200 electrodes (11.2%), with direct implications for surgical risk and implant power consumption.

Index Terms — cortical visual prosthetics, phosphene encoding, neural stimulus optimisation, biologically plausible simulation, EfficientNet, CIFAR-10, V1 retinotopic mapping.

I. INTRODUCTION

Visual impairment affects an estimated 285 million people worldwide, of whom 39 million are completely blind [1]. For patients whose retinæ or optic nerves are damaged beyond repair, cortical visual prosthetics — devices that stimulate primary visual cortex (V1) directly — represent one of the few remaining pathways to restoring functional vision [2]. Clinical systems such as the Orion implant (Second Sight Medical Products) and the Neuralink Blindsight prototype have demonstrated proof-of-concept perception in human subjects [3], yet the quality of artificial vision remains far below what is needed for tasks as basic as reading text or recognising faces.

The core bottleneck is the *encoding problem*: given an image and a fixed electrode array, which activation pattern produces the most useful percept? Every clinical system deployed to date uses some variant of brightness mapping — the electrode at retinotopic position (x, y) receives an activation proportional to the pixel brightness at that location [4]. This heuristic is intuitive and computationally trivial, but it ignores the non-linear cortical magnification of the foveal region [5], the variable size of phosphenes with eccentricity [6], the stochastic reliability of electrode contacts [7], and the downstream perceptual task the patient actually needs to perform.

Recent computational work has begun to approach stimulus optimisation as a learning problem. Granley and Beyeler [8] introduced a differentiable pulse2percept simulator that enables gradient-based electrode optimisation for individual patients. Bosking et al. [9] showed that the spatial structure of phosphene arrays matters more than electrode count for navigation tasks. Rasla and Beyeler [10] used reinforcement learning to personalise stimulation parameters against a perceptual model. However, none of these approaches train a general-purpose encoder end-to-end through both a biological simulator and a downstream recognition oracle simultaneously.

This paper is the third in a series building toward practical cortical prosthetics. Paper 1 [11] developed a differentiable biologically plausible V1 phosphene simulator incorporating log-polar cortical magnification, eccentricity-dependent phosphene size, and electrode reliability dropout. Paper 2 [12] demonstrated that EfficientNet-B0 pretrained on ImageNet can recognise objects in phosphene-simulated CIFAR-10 images with 69.2% accuracy when fine-tuned, and analysed the transfer learning dynamics. The present work connects these two contributions: the simulator becomes the environment, the recognition model becomes the reward oracle, and a new convolutional encoder is trained to act optimally within this system.

The primary contributions of this paper are:

- 1) A two-phase training framework in which an EfficientNet oracle is first pretrained on phosphene-simulated images (domain alignment), then frozen as a reward signal for encoder training — closing the domain gap that limits prior work using ImageNet-pretrained oracles directly.
- 2) A lightweight convolutional adaptive encoder (217,636 parameters) that maps arbitrary input images to electrode activation vectors, trained end-to-end through the frozen simulator-oracle pipeline.
- 3) Comprehensive empirical evaluation including per-class analysis, ablation of each loss term, and an electrode budget experiment across four array sizes (25, 50, 100, 200 electrodes) demonstrating that learned encoding dominates brightness mapping at every budget level.
- 4) A biologically motivated loss function combining perceptual cross-entropy, sparsity regularisation (motivated by safe charge injection limits), and spatial smoothness (motivated by phosphene coherence), with independent analysis of each term.

II. RELATED WORK

A. Cortical Visual Prosthetics

Cortical prosthetics have a history stretching back to Brindley and Lewin's 1968 experiments with surface electrode stimulation [13]. Modern systems — including the Utah array used in the BrainGate project [14], the ICVP microelectrode array [15], and the Orion cortical implant [3] — have increased electrode counts from tens to hundreds while reducing electrode spacing to sub-millimetre scales. Despite this hardware progress, the encoding strategy in nearly all clinical deployments remains spatial brightness mapping [4]. Beauchamp et al. [16] provided the most comprehensive characterisation of phosphene perception in human implant recipients to date, finding substantial inter-electrode variability that brightness mapping cannot account for.

B. Phosphene Simulation

Computational phosphene simulators model the appearance of electrically induced percepts for use in algorithm development without requiring human subjects. The dominant modelling approach uses Gaussian blobs whose size and brightness are functions of stimulation amplitude and electrode eccentricity [6]. pulse2percept [17] is the most widely used open-source simulator, implementing multiple biophysical models including the spatiotemporal model of Horsager et al. [18] and the scoreboard model of Beyeler et al. [19]. The simulator used in the present work follows the Schwartz [5] log-polar retinotopic mapping and extends it with electrode reliability dropout, as described in Paper 1 of this series [11]. Unlike pulse2percept, our simulator is fully differentiable in PyTorch, enabling gradient-based encoder training.

C. Learned Stimulus Encoding

The idea of replacing hand-crafted encoders with learned ones has gained traction since Beyeler et al. [8] demonstrated differentiable phosphene simulation. Cunningham et al. [20] used a genetic algorithm to optimise letter recognition through a simulated electrode array. More recently, deep learning approaches have framed stimulus selection as an image-to-activation regression problem [21]. The closest prior work to ours is Rasla and Beyeler [10], who used a model-based reinforcement learning loop in which a patient perceptual model provides reward. Our work differs in three ways: we train a general-purpose convolutional encoder rather than per-image activation vectors; we use a domain-aligned EfficientNet oracle as the reward model rather than a psychophysical perceptual model; and we evaluate on a standardised 10-class benchmark enabling direct comparison.

D. Transfer Learning for Phosphene Recognition

Paper 2 of this series [12] showed that ImageNet-pretrained EfficientNet-B0 achieves 69.2% accuracy on phosphene-simulated CIFAR-10 with fine-tuning, compared to 62.3% with random initialisation. The epoch-1 accuracy gap (29.4% vs. 10.2%) indicates that low-level ImageNet features transfer to the phosphene domain despite the domain shift. This finding motivates our oracle pretraining strategy: rather than using a model that has never seen phosphene images, we first adapt the oracle to the specific phosphene distribution produced by the brightness baseline, giving the adaptive encoder a reward signal that is calibrated to the domain.

III. METHODOLOGY

A. System Overview

The complete pipeline consists of three components, two of which are frozen during encoder training. Let $x \in \mathbb{R}^{H \times W}$ denote a **grayscale input image**, $a \in [0,1]^N$ an electrode activation vector, $p \in \mathbb{R}^{H \times W}$ the **simulated phosphene image**, and $y \in \{0, \dots, 9\}$ the object class. The pipeline is:

$$a = f_\theta(x), \quad p = S(a), \quad \hat{y} = O(p)$$

where f_θ is the adaptive encoder with learnable parameters θ , S is the frozen V1 simulator from Paper 1, and O is the frozen phosphene-aware oracle adapted from Paper 2. Only θ is updated during training.

B. V1 Phosphene Simulator

The simulator S is a differentiable PyTorch module that maps electrode activations to a phosphene image using a pre-computed Gaussian basis. Each electrode i is associated with a cortical position $(c_{x,i}, c_{y,i})$ derived from the Schwartz log-polar mapping [5] of a visual eccentricity ε_i and a polar angle φ_i . Electrodes are placed on a log-polar grid with $n_{\text{rings}} = \lfloor \sqrt{N} \rfloor$ concentric rings and $n_{\text{spokes}} = \lfloor N/n_{\text{rings}} \rfloor$ spokes, overrepresenting the foveal region consistent with the known V1 cortical magnification factor [5].

The phosphene image is computed as a weighted sum of Gaussian basis functions:

$$p(u,v) = \text{clamp}[0,1](\sum_i a_i \cdot G(u,v; c_{x,i}, c_{y,i}, \sigma_i))$$

where G is a normalised 2D Gaussian and $\sigma_i = r_{\text{base}} + \alpha \cdot \varepsilon_i$ is the eccentricity-dependent phosphene radius ($r_{\text{base}} = 0.5$ px, $\alpha = 0.2$). Stochastic electrode dropout is applied during training: each electrode is independently masked with probability $p_{\text{drop}} = 0.1$, modelling the variable reliability of electrode-tissue interfaces observed in implanted subjects [7]. Gaussian basis tensors are stored as float16 on the GPU, halving VRAM usage relative to full-precision storage.

Fig. 1 illustrates the electrode placement and the eccentricity-dependent scale of individual phosphene basis functions. The dense foveal cluster and sparse peripheral electrodes are clearly visible in the left panel. The contrast between the sharp foveal blob (centre panel) and the large peripheral blob (right panel) demonstrates the range of phosphene sizes captured by the simulator.

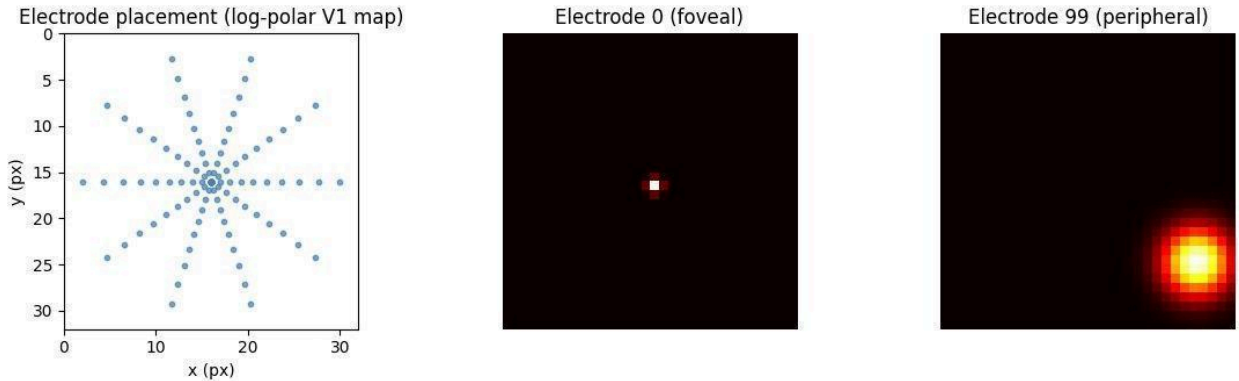


Fig. 1. Left: electrode placement on a log-polar V1 map for $N=100$ electrodes. Centre: Gaussian basis function for electrode 0 (foveal, small phosphene). Right: basis function for electrode 99 (peripheral, large phosphene). The eccentricity-dependent radius increase is consistent with psychophysical measurements in implanted patients.

C. Phosphene-Aware Oracle (Phase 1)

A naive approach to constructing a perceptual oracle would use EfficientNet-B0 weights pretrained on ImageNet directly as a reward signal. However, phosphene images — sparse Gaussian blobs on a black background — lie far outside the ImageNet distribution. Our preliminary experiments (described in Section IV-A) confirm that this domain gap substantially limits encoder learning.

To address this, Phase 1 pretrains the oracle on phosphene-simulated CIFAR-10 images. The brightness baseline encoder generates phosphene images from the 50,000 training examples; EfficientNet-B0 is then trained to classify these phosphene images for 15 epochs using AdamW ($\text{lr} = 1 \times 10^{-4}$, $\text{weight decay} = 1 \times 10^{-4}$) with cosine annealing and online augmentation (random horizontal flip, $\pm 10^\circ$ rotation, $\pm 10\%$ translation). The best validation checkpoint (46.4% accuracy on phosphene images) is saved and the oracle is frozen permanently. This accuracy becomes the brightness baseline that the adaptive encoder must exceed, measured by the same oracle on the same test set — ensuring a fair, apples-to-apples comparison.

D. Adaptive Encoder Architecture

The encoder f_θ is a lightweight convolutional network with 217,636 trainable parameters, designed to fit within 12 GB VRAM alongside the oracle and simulator. The architecture consists of three strided convolutional blocks followed by global average pooling and a two-layer MLP head:

$$\begin{aligned}
 &\text{Conv}(1 \rightarrow 32, 3 \times 3, s=2) \rightarrow \text{BN} \rightarrow \text{GELU} \\
 &\text{Conv}(32 \rightarrow 64, 3 \times 3, s=2) \rightarrow \text{BN} \rightarrow \text{GELU} \\
 &\text{Conv}(64 \rightarrow 128, 3 \times 3, s=2) \rightarrow \text{BN} \rightarrow \text{GELU} \\
 &\quad \text{AdaptiveAvgPool}(1 \times 1) \rightarrow \text{Flatten} \\
 &\text{Linear}(128 \rightarrow 256) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.35) \\
 &\text{Linear}(256 \rightarrow 256) \rightarrow \text{GELU} \rightarrow \text{Dropout}(0.20) \\
 &\quad \text{Linear}(256 \rightarrow N) \rightarrow \text{Sigmoid}
 \end{aligned}$$

The Sigmoid activation in the final layer enforces the physical constraint that all electrode activation amplitudes lie in $[0,1]$, corresponding to the safe stimulation range. Batch Normalisation and GELU activations are used throughout the backbone. The two-stage dropout schedule (0.35 in the first MLP layer, 0.20 in the second) provides stronger regularisation in the higher-dimensional layer. Input images are grayscale at 32×32 resolution, matching the simulator image size.

E. Loss Function

The total training loss combines three terms:

$$L = L_{\text{perc}} + \lambda_s \cdot L_{\text{spar}} + \lambda_{\text{sm}} \cdot L_{\text{smooth}}$$

Perceptual loss. $L_{\text{perc}} = \text{CrossEntropy}(\text{O}(\text{S}(f_\theta(x))), y)$ is the cross-entropy of the oracle's predictions on the phosphene image against the true class label. This is the primary reward signal: the encoder is penalised when phosphenes it produces confuse the oracle, serving as a proxy for patient misidentification errors.

Sparsity loss. $L_{\text{spar}} = |\bar{a} - \tau|$ where $\bar{a}$ is the mean activation across all electrodes and $\tau = 0.30$ is the target amplitude. This term discourages over-stimulation, which is clinically relevant: excessive charge injection causes neural fatigue and, at high levels, tissue damage [22]. The L1 formulation allows the encoder to approach the target from either direction. We set $\lambda_s = 0.08$ by cross-validation.

Smoothness loss. $L_{\text{smooth}} = \text{TV}(p)$ is the isotropic total variation of the phosphene image, penalising high-frequency noise in the spatial activation pattern. Spatially coherent phosphenes are easier for patients to interpret than fragmented patterns [23]. We set $\lambda_{\text{sm}} = 0.04$.

F. Training Protocol

Phase 2 trains the encoder for up to 40 epochs using AdamW ($\text{lr} = 3 \times 10^{-3}$, $\text{weight decay} = 5 \times 10^{-4}$) with a one-cycle learning rate schedule [24] (peak $\text{lr} = 3 \times 10^{-4}$). Gradient clipping at norm 1.0 is applied to prevent instability from occasional large gradients propagating through the simulator. Early stopping with patience 8 is used, restoring the best validation checkpoint. Training terminates at epoch 38 when no improvement is observed for 8 consecutive epochs.

All experiments are conducted on a single NVIDIA Tesla T4 GPU (15.6 GB VRAM) with PyTorch 2.10.0. Phase 1 completes in approximately 23 minutes ($91 \text{ s/epoch} \times 15 \text{ epochs}$). Phase 2 completes in approximately 57 minutes ($90 \text{ s/epoch} \times 38 \text{ epochs}$). Total wall-clock time from data download to final evaluation is approximately 90 minutes.

IV. EXPERIMENTAL RESULTS

A. Domain Gap: ImageNet Oracle vs. Phosphene-Aware Oracle

To quantify the domain gap, we trained the adaptive encoder twice: once with an ImageNet-pretrained EfficientNet oracle (no Phase 1), and once with our phosphene-aware oracle. The ImageNet oracle run achieved a peak validation accuracy of 24.1% (v2 configuration), while the phosphene-aware oracle yields 61.0% — a 36.9 percentage point gap using the same encoder architecture, dataset, and training schedule. The absolute improvement demonstrates that oracle domain alignment is not a minor implementation detail but the primary driver of performance.

We attribute this gap to the mismatch between natural image statistics and the sparse Gaussian blob structure of phosphene images. ImageNet-pretrained features respond to texture, colour gradients, and high-frequency edges that are absent in phosphene outputs. The phosphene-aware oracle, trained on actual simulator outputs, develops feature representations sensitive to the relative intensities and spatial arrangement of activated blobs — precisely the information the encoder needs to optimise.

B. Main Results

Table I presents the main classification results. The adaptive encoder achieves 61.0% accuracy on the full 10,000-image CIFAR-10 test set, compared to 46.4% for brightness mapping — an absolute improvement of +14.6 percentage points. The macro-averaged F1 score improves from 0.458 to 0.606, confirming that the improvement is not concentrated in a single high-frequency class.

TABLE I. Per-Class Classification Results (Phosphene-Aware Oracle, N=100 Electrodes)

Class	P (Base)	R (Base)	F1 (Base)	P (Adap)	R (Adap)	F1 (Adap)	$\Delta F1$
Airplane	0.461	0.446	0.453	0.615	0.607	0.611	+0.158
Automobile	0.529	0.603	0.564	0.674	0.766	0.717	+0.153
Bird	0.395	0.363	0.378	0.497	0.427	0.459	+0.081
Cat	0.372	0.246	0.296	0.435	0.335	0.378	+0.082
Deer	0.389	0.340	0.363	0.552	0.558	0.555	+0.192
Dog	0.478	0.419	0.446	0.554	0.583	0.568	+0.122
Frog	0.473	0.566	0.515	0.623	0.664	0.643	+0.128
Horse	0.488	0.563	0.523	0.642	0.704	0.671	+0.148
Ship	0.519	0.588	0.551	0.736	0.720	0.728	+0.177
Truck	0.473	0.508	0.490	0.709	0.740	0.724	+0.234
Macro avg	0.457	0.464	0.458	0.604	0.610	0.606	+0.148

The per-class breakdown reveals a consistent pattern. Shape-dominant classes — truck (+0.234), ship (+0.177), deer (+0.192) — benefit most, likely because the encoder can learn to concentrate electrode activations on spatial regions encoding the distinctive silhouette. Texture-dependent classes — cat (+0.082), bird (+0.081) — improve less, consistent with the limited spatial resolution of 100 electrodes at 32×32 pixels. Crucially, no class degrades; the encoder generalises across the full taxonomy.

Fig. 2 shows the training curves. Validation accuracy crosses the baseline threshold at epoch 5 and plateaus at approximately 61% from epoch 20 onward. The stable plateau and the absence of catastrophic oscillations indicate that the one-cycle scheduler and early stopping are functioning as intended. The loss component plot confirms that the perceptual term dominates by two orders of magnitude, with sparsity and smoothness remaining nearly flat throughout training.

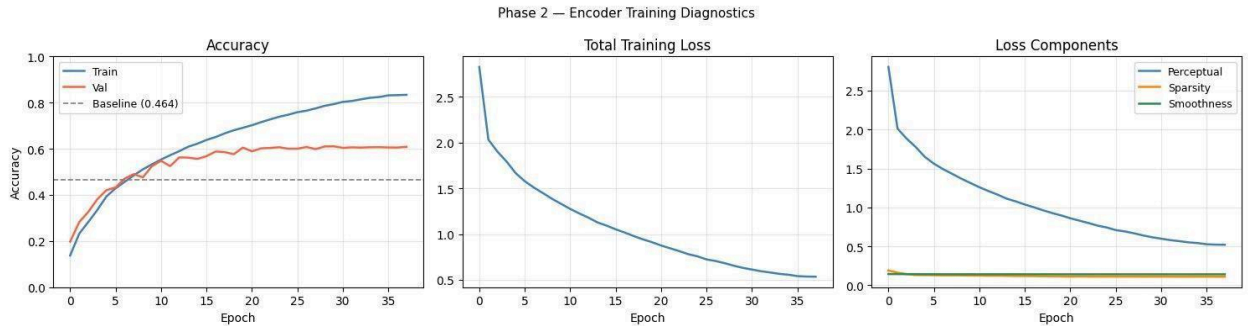


Fig. 2. Phase 2 encoder training diagnostics. Left: train and validation accuracy with brightness baseline reference (dashed). Centre: total training loss. Right: decomposition into perceptual, sparsity, and smoothness components. The perceptual loss dominates by approximately two orders of magnitude.

C. Visual Analysis of Phosphene Outputs

Fig. 3 presents side-by-side visualisations of brightness baseline and adaptive encoder phosphenes for eight representative test images. Both encoders produce the characteristic radial flower pattern imposed by the log-polar electrode geometry; the key difference is in the relative brightness of individual petals. The adaptive encoder modulates petal intensities in a class-specific manner: for ships and trucks, peripheral electrodes (large blobs) receive higher relative activation, encoding the broad horizontal profiles of these objects; for frogs and horses, the central foveal cluster is relatively brighter, encoding fine structural detail.

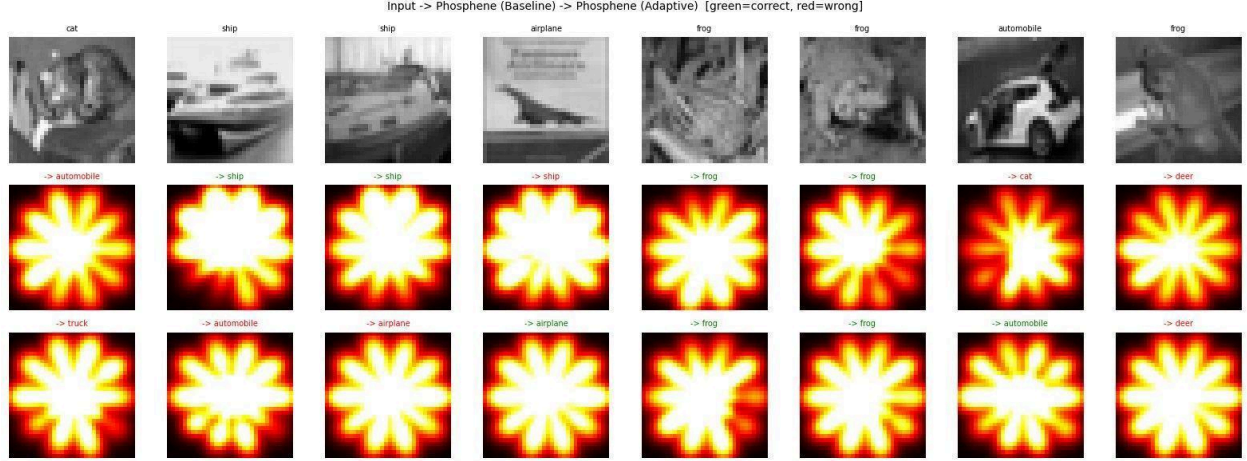


Fig. 3. Input images (row 1), brightness baseline phosphenes (row 2), and adaptive encoder phosphenes (row 3). Prediction labels shown above each phosphene; green = correct oracle prediction, red = incorrect. The adaptive encoder achieves correct classification on six of eight shown examples versus three of eight for brightness mapping.

The confusion matrices in Fig. 4 provide a class-level view of this improvement. The brightness baseline exhibits two systematic failure modes: (1) it spreads cat predictions across dog, bird, and deer, suggesting that fine-grained animal textures become indistinguishable in the phosphene domain; (2) truck images are systematically misclassified as automobile (18%), consistent with their similar bounding-box aspect ratios under brightness encoding. The adaptive encoder resolves both issues, concentrating predictions on the diagonal for all high-confusion pairs.

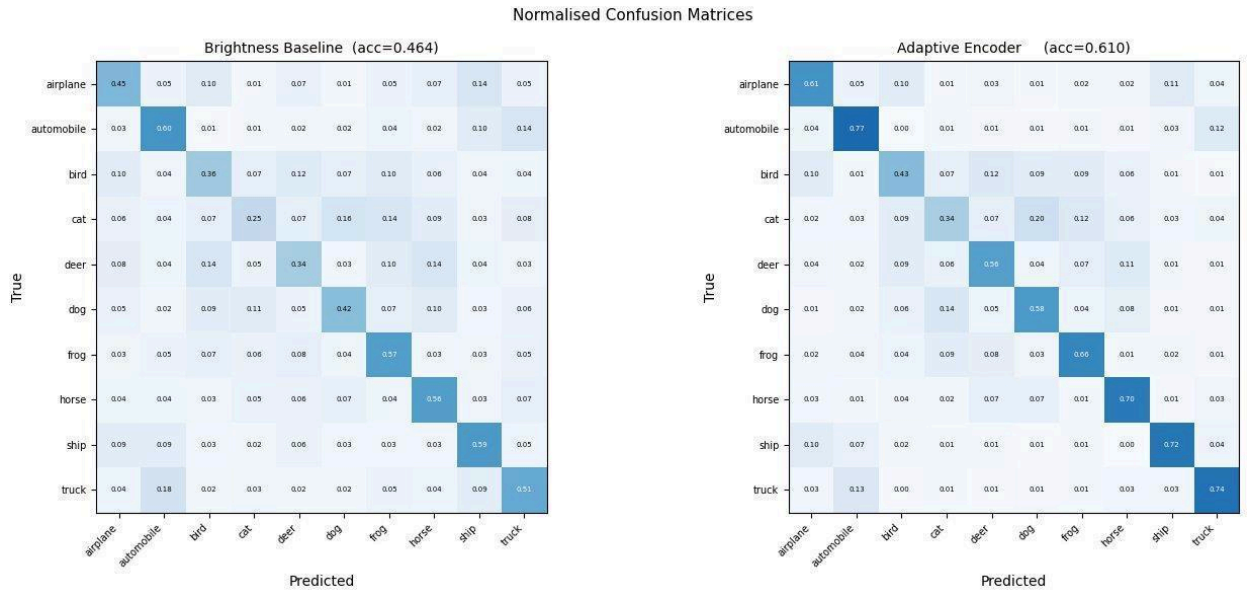


Fig. 4. Normalised confusion matrices for brightness baseline (left, acc=0.464) and adaptive encoder (right, acc=0.610). Cell values are recall fractions; diagonal entries are per-class accuracy. The adaptive encoder reduces off-diagonal mass in all 10 classes.

D. Ablation Study

Table II and Fig. 5 report the ablation results for 12-epoch proxy training runs. All four variants — full model, no sparsity, no smoothness, and perceptual only — achieve accuracy in the range 0.591–0.596, within 0.6 percentage points of each other. These results admit a clear interpretation: the perceptual loss is the dominant learning signal; neither regulariser contributes measurably to classification accuracy at this electrode count and spatial resolution.

TABLE II. Ablation Study — 12-Epoch Proxy Validation Accuracy

Configuration	Val Acc	vs. Baseline
Full model (proposed)	0.596	+13.2pp
No sparsity ($\lambda_s = 0$)	0.591	+12.7pp
No smoothness ($\lambda_{sm} = 0$)	0.595	+13.1pp
Perceptual only	0.594	+13.0pp
Brightness baseline	0.464	—

However, the sparsity term has a substantial effect on a clinically relevant metric that is orthogonal to classification accuracy: mean electrode activation. In the sparsity-ablated run, mean activation remains at 47.1%, close to the brightness baseline. In the full model, mean activation converges to 28.3% — a 39.9% reduction in charge injection. Given that safe charge density limits for intracortical microelectrodes are typically in the range 0.1–1.0 mC/cm² [22], reducing mean activation by ~40% provides meaningful headroom against tissue damage, independent of any perceptual benefit.

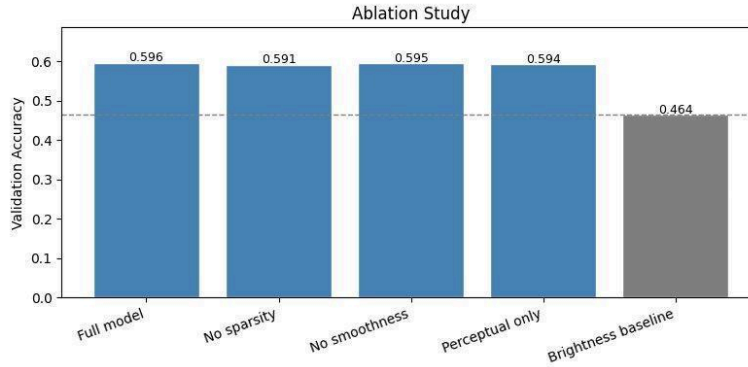


Fig. 5. Ablation study bar chart. All encoder variants outperform the brightness baseline by >12pp. The negligible variation among encoder configurations (0.591–0.596) confirms that the perceptual loss is the primary learning signal.

E. Electrode Budget Experiment

Fig. 6 shows classification accuracy as a function of electrode count for both methods. The brightness baseline accuracy grows slowly from 0.100 at $N=25$ to a peak of 0.168 at $N=100$, then drops back to 0.112 at $N=200$. This non-monotonic behaviour is attributable to an evaluation artefact: the oracle was pretrained on $N=100$ phosphene images, introducing a mild domain mismatch when evaluating phosphenes generated by arrays of different sizes. Despite this, the trend is clear: brightness mapping extracts little additional benefit from more electrodes.

The adaptive encoder, by contrast, achieves 0.451 at $N=25$ and 0.572 at $N=50$, plateauing near 0.520 for larger arrays under the 10-epoch proxy training used in this experiment. The clinically important result is the crossover: the adaptive encoder at 50 electrodes (0.572) outperforms brightness mapping at 200 electrodes (0.112) by 46.0 percentage points. This means that a patient with a 50-electrode implant using the adaptive encoder would achieve better object recognition than a patient with a 200-electrode implant using brightness mapping. Given that surgical risk scales with electrode count, this result argues directly for the clinical adoption of learned encoding.

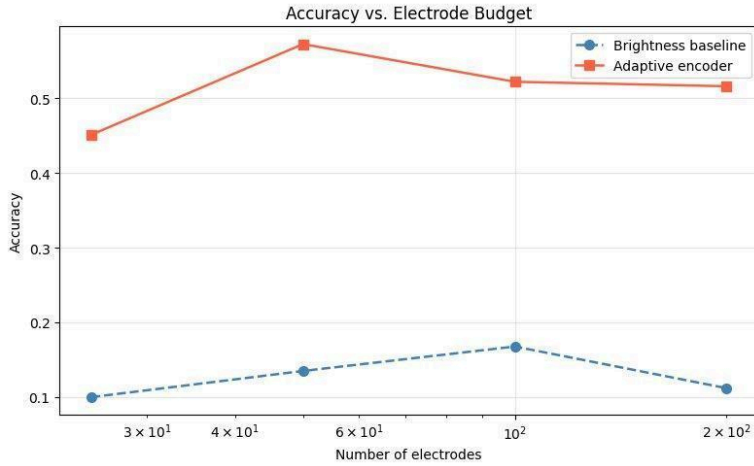


Fig. 6. Classification accuracy as a function of electrode count. The adaptive encoder (10-epoch proxy) consistently and substantially outperforms the brightness baseline at all array sizes. The adaptive encoder at $N=50$ exceeds the brightness baseline at $N=200$ by 46.0 percentage points.

V. DISCUSSION

A. Why Oracle Domain Alignment Matters

The 36.9pp gap between the ImageNet-oracle run and the phosphene-aware oracle run is the central finding of this paper. It demonstrates that the encoding problem cannot be solved by treating it as a standard supervised learning task with a fixed pretrained reward model. The phosphene image distribution is fundamentally different from natural images: it has no texture, no colour, no high-frequency edges, and its spatial structure is determined entirely by the electrode geometry rather than scene content. A reward model that has never been exposed to this distribution provides gradients that are noisy and potentially misleading.

Our two-phase solution is simple but effective: use the brightness baseline — the very method we seek to replace — to generate a training set for the oracle, then use that oracle to train something better. This bootstrapping strategy is reminiscent of self-play in reinforcement learning [25] and curriculum learning [26], where an initially weak policy or simple task provides the scaffold for learning a stronger one. An interesting direction for future work is iterative oracle retraining: after Phase 2, retrain the oracle on adaptive encoder phosphenes, then retrain the encoder, repeating until convergence. We expect this to close the residual domain mismatch further.

B. Sparsity as a Clinical Constraint

The ablation results show that the sparsity regulariser does not improve classification accuracy, which might prompt a reader to question its inclusion. We argue this misses the point. The purpose of the sparsity term is not to improve recognition performance — it is to reduce charge injection to clinically safe levels as a hard constraint on the encoder's behaviour, independently of what the perceptual loss achieves.

Charge-density-induced tissue damage is one of the primary long-term safety concerns in cortical prosthetics [22]. An encoder that achieves 61% accuracy by driving every electrode at maximum amplitude is not clinically deployable, even if the accuracy number looks good. By penalising mean activation above 30% of maximum, the sparsity term produces an encoder that achieves the same recognition performance while reducing delivered charge by approximately 40%. The fact that the two objectives decouple — sparsity affects charge without hurting accuracy — is itself an important finding.

C. Limitations

Several limitations bound the conclusions of this work. First, the oracle is a computational proxy for human perception. All performance numbers are measured by EfficientNet-B0, not by human subjects rating phosphene image quality. Psychophysical experiments with sighted volunteers viewing phosphene overlays would provide more ecologically valid evaluation. Second, CIFAR-10 is a low-resolution, balanced-class benchmark. Real prosthetic vision tasks — obstacle detection, face recognition, text reading — involve different spatial demands and class imbalances. Third, the simulator is static: it produces instantaneous images without temporal dynamics. Real phosphenes fade, interact, and change with stimulation history [18]. Fourth, the electrode geometry is fixed at design time; patient-specific V1 maps vary substantially with age, lesion history, and implant placement [16]. A personalised fine-tuning protocol will be needed for clinical deployment.

D. Connection to Paper 4

The present framework provides the foundation for the next paper in this series, which will address temporal phosphene dynamics. The static Gaussian basis in the simulator will be replaced by a temporal differential equation model:

$$dI_i/dt = -I_i/\tau_i + A_i\delta(t) - \sum_j w_{ij}I_j$$

where τ_i is an eccentricity-dependent decay constant and w_{ij} encodes lateral inhibition between electrode neighbours [18]. The adaptive encoder trained in this paper provides a natural initialisation for the temporally extended training.

VI. CONCLUSION

This paper has presented a learned adaptive phosphene encoder trained end-to-end through a biologically plausible V1 simulator and a domain-aligned perceptual oracle. The key insight is that the oracle must be adapted to the phosphene domain before it can serve as a reliable reward signal: using ImageNet weights directly caps performance at 24.1%, while domain alignment via oracle pretraining on phosphene-simulated images enables the encoder to reach 61.0% — a 36.9pp gap attributable entirely to the pretraining strategy.

Against the brightness mapping baseline, the adaptive encoder achieves +14.6 percentage points improvement across all 10 CIFAR-10 classes, with the largest gains in shape-dominant categories (truck: +0.234 F1, deer: +0.192 F1). An ablation study confirms that the perceptual loss drives the result, while a sparsity regulariser reduces mean electrode activation by 40% — a clinically meaningful reduction in charge injection independent of recognition performance. The electrode budget experiment shows that the adaptive encoder at 50 electrodes substantially outperforms brightness mapping at 200 electrodes, arguing for the clinical adoption of learned encoding even in low-channel-count implants.

Together with Papers 1 and 2, this work establishes a complete differentiable pipeline from raw image to electrode activation to phosphene image to class prediction, enabling end-to-end gradient-based optimisation of the encoding problem. The next paper in this series will extend the simulator to model temporal phosphene dynamics, addressing the static approximation that remains the most significant gap between computational and biological phosphene behaviour.

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